

# 1. Arrays

## ● Easy

### Q1. Find Maximum and Minimum

Write a C program to find the largest and smallest elements in an integer array.

Example:

Input: [5, 2, 9, 1, 7]  
Output: Maximum = 9  
Minimum = 1

### Q2. Count Even and Odd Numbers

Given an integer array, count how many elements are even and how many are odd.

Example:

Input: [2, 7, 4, 9, 6, 3]  
Output: Even = 3  
Odd = 3

## ● Moderate

### Q3. Remove Duplicates

Given an array, remove duplicate elements and print only the unique elements.

Example:

Input: [2, 3, 2, 5, 3, 7]  
Output: [2, 3, 5, 7]

### Q4. Find the Second Largest Element

Find the second-largest **distinct** element in an array without sorting the array.

Example:

Input: [10, 5, 8, 10, 3]  
Output: 8

### Q5. Maximum Subarray Sum

Find the contiguous subarray whose elements have the maximum sum.

Example:

Input: [-2, 1, -3, 4, -1, 2, 1, -5, 4]  
Output: 6

---

## 2. Structures

### ● Easy

#### Q1. Student Details

Create a structure `Student` containing:

- Roll number
- Name
- Marks

Read details of one student and display them.

---

#### Q2. Employee Salary

Create a structure `Employee` containing:

- Employee ID
- Name
- Basic salary

Calculate and display the employee's annual salary.

---

### ● Moderate

#### Q3. Find Student with Highest Marks

Create a structure `Student` containing:

```
rollNo
name
marks
```

Store details of N students and find the student who scored the highest marks.

Example:

Input:

```
101 Arun 78
102 Ravi 91
103 Kumar 85
```

Output:

```
Highest Marks: Ravi
Marks: 91
```

---

#### Q4. Sort Employees by Salary

Create an `Employee` structure containing:

```
id
name
salary
```

Store details of N employees and sort them in **ascending order of salary**.

Example:

Input:

```
101 A 50000
102 B 35000
103 C 45000
```

Output:

```
102 B 35000
103 C 45000
101 A 50000
```

---

#### Q5. Nested Structure – Student Address

Create two structures:

```
struct Address {
    char city[30];
    int pincode;
};

struct Student {
    int rollNo;
    char name[30];
    struct Address address;
};
```

Read and display the details of N students. Then print the students who belong to a given city.

---

## 3. Pointers

● Easy

#### Q1. Swap Two Numbers Using Pointers

Write a C program to swap two integers using pointers.

Example:

```
Before: a = 10, b = 20
After:  a = 20, b = 10
```

---

## Q2. Find Maximum Using Pointers

Write a function that accepts two integers using pointers and returns the larger value.

Example:

Input: 10 25

Output: 25

---

## 🕒 Moderate

### Q3. Reverse an Array Using Pointers

Write a program to reverse an integer array using **only pointer arithmetic**.

Example:

Input: [1, 2, 3, 4, 5]

Output: [5, 4, 3, 2, 1]

Try to avoid using array indexing such as:

`arr[i]`

---

### Q4. Count Vowels Using a Pointer

Write a function that accepts a string using a character pointer and counts the number of vowels.

Example:

Input: "programming"

Output: 3

Use pointer traversal instead of:

`str[i]`

---

### Q5. Dynamic Array Using `malloc()`

Use `malloc()` to dynamically allocate memory for N integers.

Then:

1. Read the elements.
2. Find the maximum and minimum.
3. Calculate the average.
4. Free the allocated memory using `free()`.

Example:

Input: 5

10 20 5 40 15

Output:

Maximum = 40

Minimum = 5

Average = 18

---